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SLUMS AS SITES OF ENVIRONMENTAL (IN)JUSTICE

Individual Report

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Submitted to

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1. Executive Summary

This report assesses digital adaptation in India using the audited all-India estimates of the Comprehensive Modular Survey: Telecom (CMS-T), fielded in January–March 2025 as part of the NSS 80th Round (NSO, 2025). The survey covered 4,382 first-stage units, 34,950 households and 142,065 persons. Adaptation is treated as three linked dimensions, namely access, usage and self-reported functional capability, and is examined by gender, rural–urban sector and age band. Education, income, social-group and State/UT breakdowns are not in the published tables and are therefore not analysed.

Principal findings

- **Household connectivity is near-universal but mobile-based.** 95.2% of households possess a mobile phone, 85.5% at least one smartphone and 86.3% report internet facility in the premises; yet 84.3% of internet households use a mobile network only and just 7.1% of all households have optical fibre.
- **Individual adaptation is lower and more uneven.** Among persons aged 15+, 85.7% used a mobile phone, 70.2% own one and 70.0% used the internet in the last three months; self-reported task capability ranges from 63.6% (sent an attachment) to 13.1% (word processing).
- **Youth have converged on use, not on ownership or capability.** 93.6% of 15–29-year-olds used the internet, but the gender gaps in ownership (20.3 pp) and online-banking ability (21.8 pp) remain above 20 points.
- **Gender gaps are widest in ownership.** Men 83.9% versus women 56.2% (27.7 pp), followed by online banking (23.5 pp). Every gap for which a relative standard error is published is statistically distinguishable from zero.
- **Rural–urban gaps grow with task complexity.** Rural–urban gaps are 8.1 pp for mobile use but 19.4 pp for e-mail and 20.4 pp for online banking; rural adults report word-processing ability at 8.3% against 22.7% in urban areas.
- **Disadvantages compound.** Rural women's ownership (48.4%) trails urban men's (90.0%) by 41.6 pp.
- **Change since 2022–23 is large but descriptive.** Internet use among persons 15+ rose from 57.5% to 70.0%; the survey designs differ and no significance test is possible.

Implication. Digital adaptation in India has moved beyond the connection stage; the binding constraints are now personal device ownership and functional capability, concentrated among women and rural residents. These are interpretations of cross-sectional, self-reported, published aggregates and are not causal estimates.

2. Introduction

Policy and public debate on India's digital expansion is often framed through connection statistics, yet a connection is not use and use is not capability. A household may report an internet facility while one member controls the only handset; a young person may use the internet daily yet be unable to complete a banking transaction or report online fraud. Rural digital-literacy programmes illustrate the policy emphasis on capability: the government reported that 6.39 crore rural people had been trained under the Pradhan Mantri Gramin Digital Saksharta Abhiyan as on 31 March 2024 against a target of six crore households (IANS, 2024).

The CMS-T is the first National Statistics Office (NSO) survey dedicated to telecom use and gives a rare opportunity to see household access, individual ownership, usage and capability in one frame. This report asks four questions:

- **RQ1.** How far has adaptation progressed across access, usage and functional capability?
- **RQ2.** How do gaps differ by gender, rural–urban sector and age band?
- **RQ3.** Do disadvantages compound, for example for rural women?
- **RQ4.** What has changed since the 2022–23 Comprehensive Annual Modular Survey (CAMS), where comparison is defensible?

The report follows the data. Where the brief asks for education, socio-economic or regional analysis, the published telecom tables do not support it (Section 4.5); 'region' is therefore operationalised as the rural–urban sector, and State/UT comparisons are excluded because the survey was not designed for them. Sections 3 and 4 set out the literature and methods, Section 5 presents the evidence, and Sections 6–9 discuss, recommend, qualify and conclude.

3. Literature Review and Conceptual Framework

From access to inequality in use. Early debates treated the digital divide as a binary of connected versus unconnected. DiMaggio and Hargittai (2001) argued that, as penetration grows, attention should shift to digital inequality among those already online, identifying five dimensions: equipment, autonomy of use, skill, social support and purpose. Hargittai (2002) showed that users differ widely in online skills. Van Dijk (2005) distinguished motivational, material, skills and usage access, and Scheerder et al. (2017), reviewing second- and third-level divides, found that the determinants of skills and uses are largely sociodemographic and that outcomes remain under-studied. This three-level logic of access, skills and use, and outcomes underpins the access–usage–capability structure adopted here.

Material access. Van Deursen and van Dijk (2019) found that, even where 98% of a population uses the internet, inequalities persist in material access, meaning the range and quality of devices, and that smartphone-only users may form a 'mobile underclass' because mobile devices are not equivalent to computers for breadth of use. This is directly relevant where household internet is almost entirely mobile-network access. Helsper's (2012) corresponding-fields model adds that economic, cultural, social and personal resources offline map onto exclusion online, which motivates reading gender and sector gaps as resource inequalities.

Development theory. Sen's (1999) capability approach and Kleine's (2013) application to ICTs hold that technology matters to the extent that people can convert it into valued functionings; owning a phone and being able to bank, communicate and seek redress with it are different achievements. Rogers' (2003) diffusion theory predicts an S-shaped adoption curve in which gaps between groups are widest at mid-levels of adoption and narrow towards saturation, an expectation examined descriptively in Section 5.9.

The International Telecommunication Union's (ITU, n.d.) 'universal and meaningful connectivity' framework adds six dimensions (quality, availability, affordability, devices, skills and security), of which CMS-T measures devices, skills and, through cyber-fraud reporting, a security proxy.

Evidence for India. Barboni et al. (2018) reported that about 67% of men but 33% of women in their Indian sample owned phones and warned that the gap can deepen inequalities in earnings, networks and information. Using NFHS-5 (2019–21) for women aged 15–49, Roy et al. (2026) found a 25.72-point rural–urban gap in ever having used the internet, with 85.34% of it statistically attributable to observed covariates: household wealth (45.75%), household mobile ownership (27.28%) and women's education (21.61%). Their decomposition is not causal and uses a different population and outcome, so levels are not comparable with CMS-T. Globally, GSMA (2025) reports that women in low- and middle-income countries are 14% less likely than men to use mobile internet, with South Asia's gap widest at 32%; these are relative, not percentage-point, gaps.

Hypotheses. The literature suggests four expectations tested descriptively below: (H1) gaps are larger for ownership and capability than for simple use; (H2) gaps are smaller among the young; (H3) gender and sector disadvantages compound; (H4) gap size varies with adoption level, narrowing near saturation.

4. Data and Methodology

4.1 Data source and sample

The 'clean dataset' is the audited workbook of published all-India estimates transcribed from NSO (2025), Appendix A, with page references. Its source sheets (A_Sources, B_Evidence, C_Data, C2_Household, C3_Distributions, D_Comparability, E_Reliability, F_Gaps, G_Literature) were all examined. The workbook records that Tables 1, 2, 3, 6 and 7 match the supplied State/UT Excel files in all 555 State/UT × age rows. The comparator is the 2022–23 CAMS (NSO, 2024). Table 1 summarises the sample.

Table 1. Survey design and sample characteristics

Feature	CMS: Telecom 2025 (NSS 80th Round)	CAMS 2022–23 (NSS 79th Round)
Reference period	January–March 2025 (one quarter)	July 2022–June 2023 (four sub-rounds)
First-stage units (FSUs)	4,382 (rural 2,395; urban 1,987)	15,298
Households surveyed	34,950	302,086
Persons enumerated	142,065 (rural 82,573; urban 59,492)	1,299,988
Selection within FSU	8 households by SRSWOR; stratified multi-stage	20 households (5 hospitalised-case, 15 others)
Coverage	All India except inaccessible A&N villages; no State participation	All India except some A&N villages; State participation (except A&N, DNH&DD)
Skill-question universe	Persons 15+ who used a smartphone, other mobile or computer in last 3 months	Persons 15+ able to use a smartphone or computer
Collection mode	Informant-reported; no skill tests	Informant-reported; no skill tests

Source: NSO (2025, Highlights p. iv; pp. 2–3); NSO (2024, pp. iii, 1–2); audited workbook sheet D_Comparability. FSU = first-stage unit; SRSWOR = simple random sampling without replacement.

4.2 Variables and denominators

Table 2 defines the indicators. Household indicators are shares of households and individual indicators are shares of persons in the age group; 'possession' is used for households and 'ownership' for individuals,

since individual ownership requires a phone with at least one active SIM for personal use (NSO, 2025, Appendix B). Distribution tables (device, network, reasons) have owners, users or non-internet households as denominators and are never compared with all-person shares.

Table 2. Indicator definitions, dimensions and denominators

ID	Indicator (dimension)	Definition / reference period	Denominator
H1	Household mobile possession (Access)	Mobile phone, incl. smartphone, possessed for household use; date of survey	All households
H2	Household smartphone (Access)	At least one smartphone (smartphone-only + both); derived	All households
H3	Household internet (Access)	Internet facility within premises; includes mobile network	All households
H4	Household online purchase (Usage)	Goods bought via e-commerce; last 30 days	All households
A6	Mobile ownership (Access)	Phone with active SIM for personal use; date of survey	Persons in age group
U1	Mobile use (Usage)	Used a mobile at least once; last 3 months	Persons in age group
U3	Internet use (Usage)	Used internet, any device/location/purpose; last 3 months	Persons in age group
C1	Sent attachment (Capability)	Sent message with attached file; executed, last 3 months	Persons in age group*
C2	Copy-and-paste (Capability)	Copied/moved data or documents; executed, last 3 months	Persons in age group*
C3	Created presentation (Capability)	Created e-presentation; executed, last 3 months	Persons in age group*
C4	Word processing (Capability)	Able to create documents; ability, date of survey	Persons in age group*
C5	E-mail (Capability)	Able to send/receive e-mail; ability	Persons in age group*
C6	Online banking (Capability)	Able to bank online (UPI, net banking); ability	Persons in age group*
C8	Cyber-fraud reporting (Capability)	Able to complain on the cybercrime portal; ability	Persons in age group*

Source: NSO (2025), Appendix B and Schedule Block 4–5; audited workbook sheet B_Evidence. *Published as "percentage of persons", but the schedule asks skill items only of 15+ users of a mobile or computer; the base is not stated (workbook gap G01). Age bands: 15–24, 15–29, 15 and above.

4.3 Validation, weights and missing data

NSO estimates are design-weighted before publication and unit-level records were not supplied, so weights cannot be re-applied. They were instead audited by a pseudo-weight check. If an All-person estimate equals a weighted mean of the male and female (or rural and urban) estimates, the implied population share can be recovered. Across the ten individual indicators aged 15+, the implied male share is 0.500–0.509 and the implied rural share 0.654–0.667 (standard deviation 0.004), the same for access, usage and skill indicators. This is consistent with a common all-persons base for the skill indicators, though it does not prove it.

Thirteen validation checks were run (Appendix, Table 11). All 270 cells of the individual-indicator grid are present and within 0–100; every Person estimate lies between the male and female estimates, and every All-India estimate between rural and urban; gaps recompute exactly; distribution rows sum to 100

within rounding. One check failed: household network types (mobile, fixed/WiFi, both) account for 99.1–99.4% of internet households, leaving 0.6–0.9 points unexplained, so 'fixed-only' access is not reported. RSEs are published for five indicators only.

4.4 Statistical methods

Gaps are percentage-point differences from rounded published values (male minus female; urban minus rural; error up to ±0.1 pp). Because absolute gaps are bounded by prevalence, parity ratios (women/men; rural/urban) and odds ratios complement them. Where RSEs exist (mobile use, ownership, internet use, copy-paste), SE = RSE × estimate/100 and the SE of a gap is $\sqrt{(SE_1^2 + SE_2^2)}$, assuming independence; this is conservative for gender gaps because men and women come from the same clusters. Approximate 95% margins are 1.96 × SE. Compounding is measured by the interaction contrast: the rural gender gap minus the urban gender gap. The association between adoption level and gap size uses Spearman's ρ across ten indicators (n = 10, not independent, so exploratory). Comparisons with 2022–23 are descriptive. Associations are not causal.

4.5 What the data allow

The published telecom tables break indicators down by sex, sector and three age bands only. Education, household consumption, religion or social group and disability were collected in the schedule but are not tabulated; sample sizes by age within a State/UT are unknown; and the smallest States have very few units (Ladakh has two rural FSUs; Chandigarh has no rural sample; Table 12). The analysis therefore covers gender, sector and age; 'regional' differences are rural–urban; and education and socio-economic gradients are discussed only through external evidence (Roy et al., 2026).

5. Empirical Analysis

5.1 Overall digital adaptation

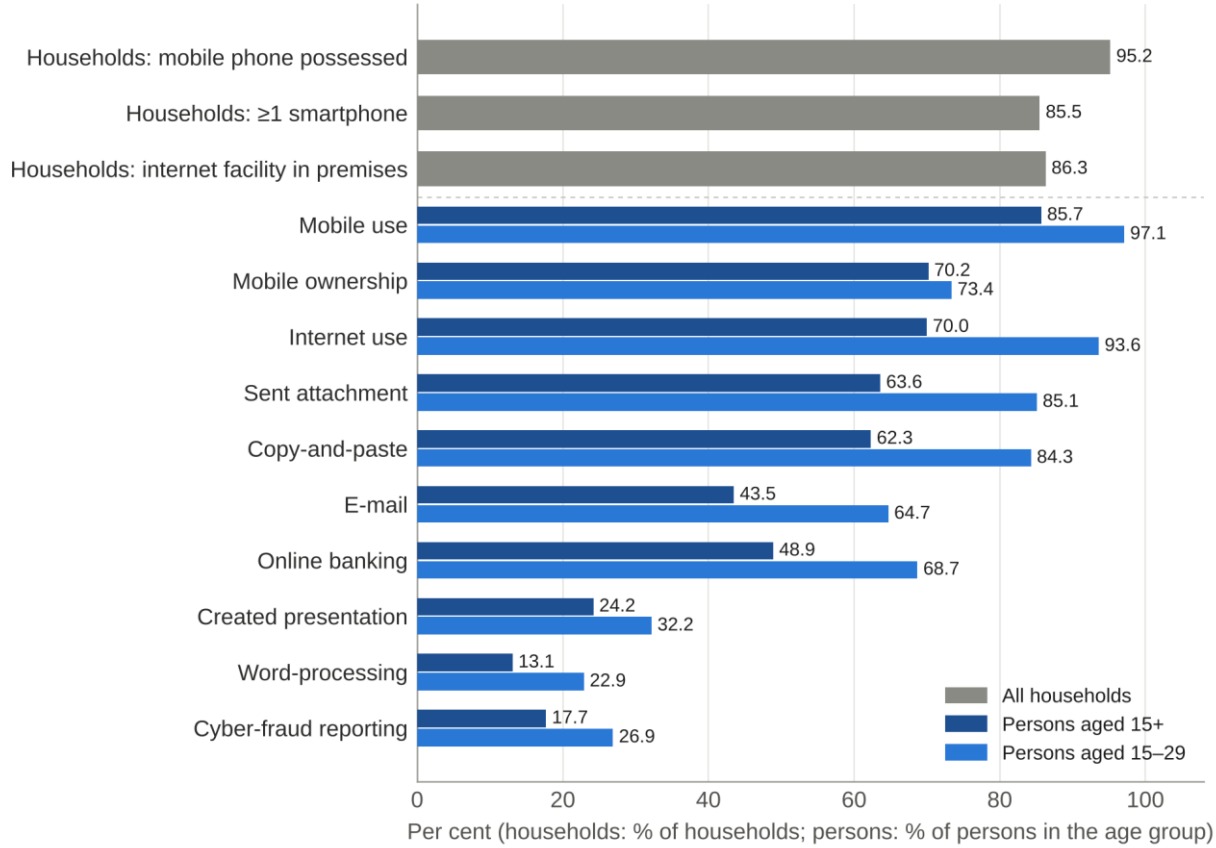
Household connectivity is high: 95.2% of households possess a mobile phone, 85.5% at least one smartphone and 86.3% report internet facility in the premises (Table 3, Figure 1). Individual measures are lower and more dispersed. Among persons aged 15+, 85.7% used a mobile and 70.0% used the internet in the last three months, but 70.2% own a phone. The 15.5-point difference between use and ownership is consistent with shared or borrowed phones, which the tables cannot separate. Capability falls from 63.6% (attachment) and 62.3% (copy-paste) to 48.9% (online banking), 43.5% (e-mail), 24.2% (presentation), 17.7% (cyber-fraud reporting) and 13.1% (word processing). These are not a strict ladder: presentation creation (execution) exceeds word-processing ability, which may reflect respondents counting phone-based tools. Against indicative ITU 2030 targets (ITU, 2022), ownership and internet use among 15+ fall short by 29.8 and 30.0 points and household internet by 13.7 points, although definitions are not harmonised.

Table 3. Overall levels of digital adaptation, all-India, 2025 (per cent)

Indicator	Dimension	Base	All persons 15+ / all households	All persons 15–29	Shortfall to ITU target (pp)
Mobile phone possession	Access	Households	95.2	–	–
≥1 smartphone	Access	Households	85.5	–	–
Internet facility in premises	Access	Households	86.3	–	13.7
Mobile use (last 3 months)	Usage	Persons	85.7	97.1	–
Mobile phone ownership	Access	Persons	70.2	73.4	29.8
Internet use (last 3 months)	Usage	Persons	70.0	93.6	30.0
Sent message with attachment	Capability	Persons	63.6	85.1	–
Copy-and-paste	Capability	Persons	62.3	84.3	–
E-mail ability	Capability	Persons	43.5	64.7	–
Online-banking ability	Capability	Persons	48.9	68.7	–
Created e-presentation	Capability	Persons	24.2	32.2	–
Word-processing ability	Capability	Persons	13.1	22.9	–
Cyber-fraud reporting ability	Capability	Persons	17.7	26.9	–

Source: NSO (2025), Tables 1–3, 6–14; ITU (2022) targets (mirror copy; to be confirmed on itu.int); author's calculations. Household smartphone share derived from Table 13 components. ITU targets: 100% for ownership, internet use and household internet. Capability rows mix execution and ability concepts (Table 2).

Figure 1. Digital adaptation by dimension: households and persons aged 15+ and 15–29, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.
Note: household smartphone share (85.5%) and mobile possession (95.2%) are derived from published Table 13 components. Skill indicators mix execution (last 3 months) and ability (date of survey).

Key finding. Household access (86.3–95.2%) sits well above individual ownership (70.2%) and most capability indicators. Among 15–29-year-olds, use is near-universal but capability tasks fall to 22.9–68.7%. **Implication.** Household connectivity overstates individual adaptation, and use overstates capability.

5.2 Household access

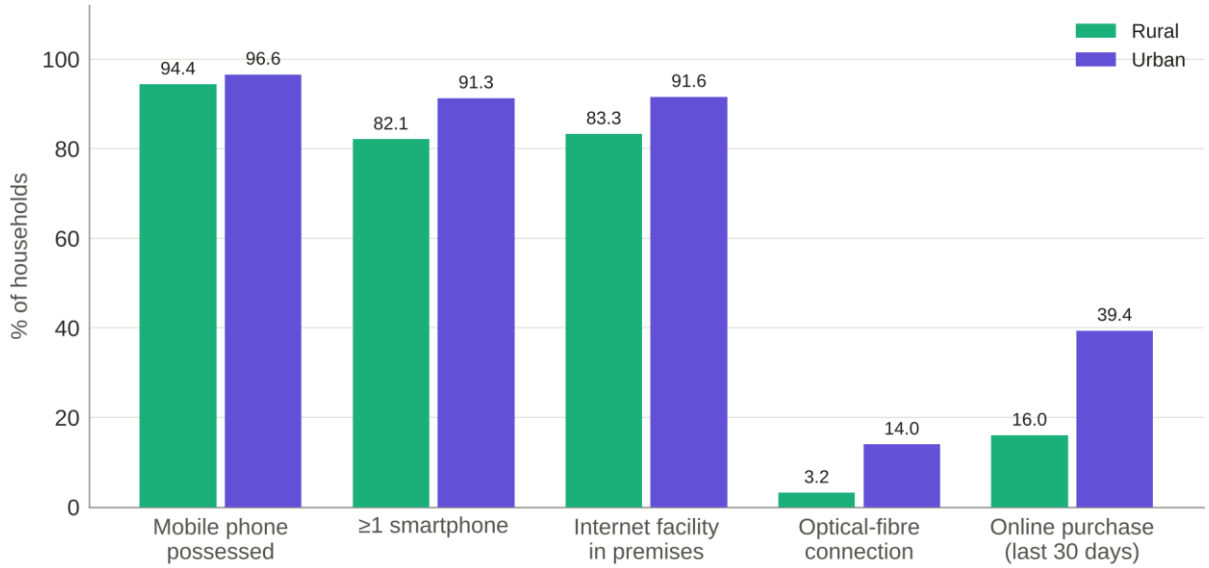
Rural–urban differences in households are small for basic possession (2.2 pp) but widen with quality and use: 9.2 pp for smartphones, 8.3 pp for internet facility, 10.8 pp for fibre and 23.4 pp for online purchases (Table 4, Figure 2). Household internet is overwhelmingly mobile: 98.8% of internet households use a mobile network, 84.3% use it exclusively, and only 14.9% have fixed or WiFi access (rural 9.1%, urban 24.0%). Among the 13.7% of households without internet, the leading stated reason is not knowing how to use it (39.2%; Figure 3).

Table 4. Household access by sector, 2025 (per cent of households)

Indicator	Rural	Urban	All-India	Urban–rural gap (pp)
Mobile phone possession	94.4	96.6	95.2	2.2
≥1 smartphone	82.1	91.3	85.5	9.2
Internet facility in premises	83.3	91.6	86.3	8.3
Optical-fibre internet	3.2	14.0	7.1	10.8
Online purchase, last 30 days	16.0	39.4	24.5	23.4
Internet households: fixed/WiFi (% of internet households)	9.1	24.0	14.9	14.9
Internet households: mobile network only (derived)	90.0	75.4	84.3	-14.6

Source: NSO (2025), Tables 13–15; Statement 14. Rows 1–5 are shares of all households; rows 6–7 are shares of households with internet. Mobile-only = mobile-network share minus "both"; mobile, fixed and "both" cover 99.1–99.4% of internet households (Table 11).

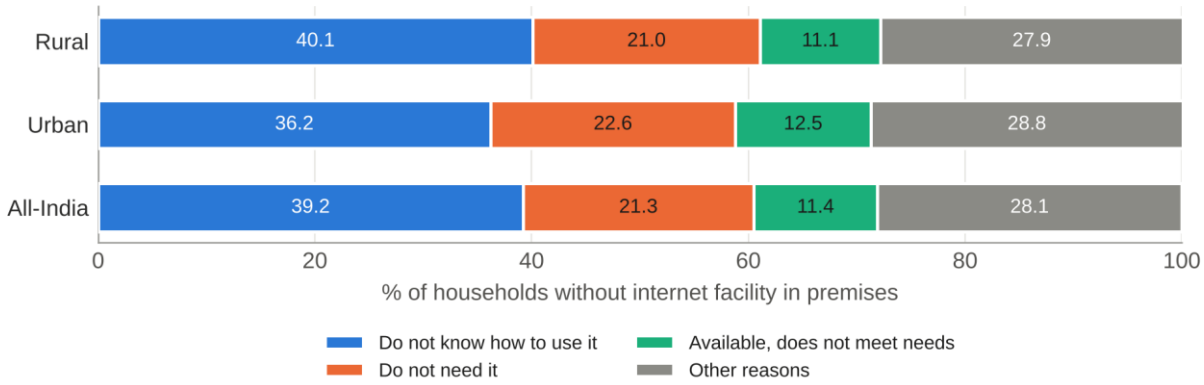
Figure 2. Household access, rural versus urban, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.
Note: denominators are all households; online purchase = goods bought via e-commerce in last 30 days.

Key finding. Possession is near-saturated in both sectors (94.4% and 96.6%), but urban households lead on smartphones, fibre (14.0% versus 3.2%) and online purchasing (39.4% versus 16.0%). **Implication.** Gaps emerge in quality and use, not in basic possession.

Figure 3. Stated major reason for no household internet facility, India, 2025



Source: NSO (2025), Table 14.1 and Fig. 11.1; author's calculations. Households without internet: rural 16.7%, urban 8.4%, all-India 13.7%.
Note: distributions sum to 100.0–100.1 because of rounding in the source.

Key finding. Not knowing how to use the internet is the largest stated reason (39.2%), ahead of not needing it (21.3%); sector differences are small. **Implication.** Skills and perceived relevance, not only infrastructure, keep households offline. Affordability is not a coded reason, so it cannot be assessed.

5.3 Gender gaps

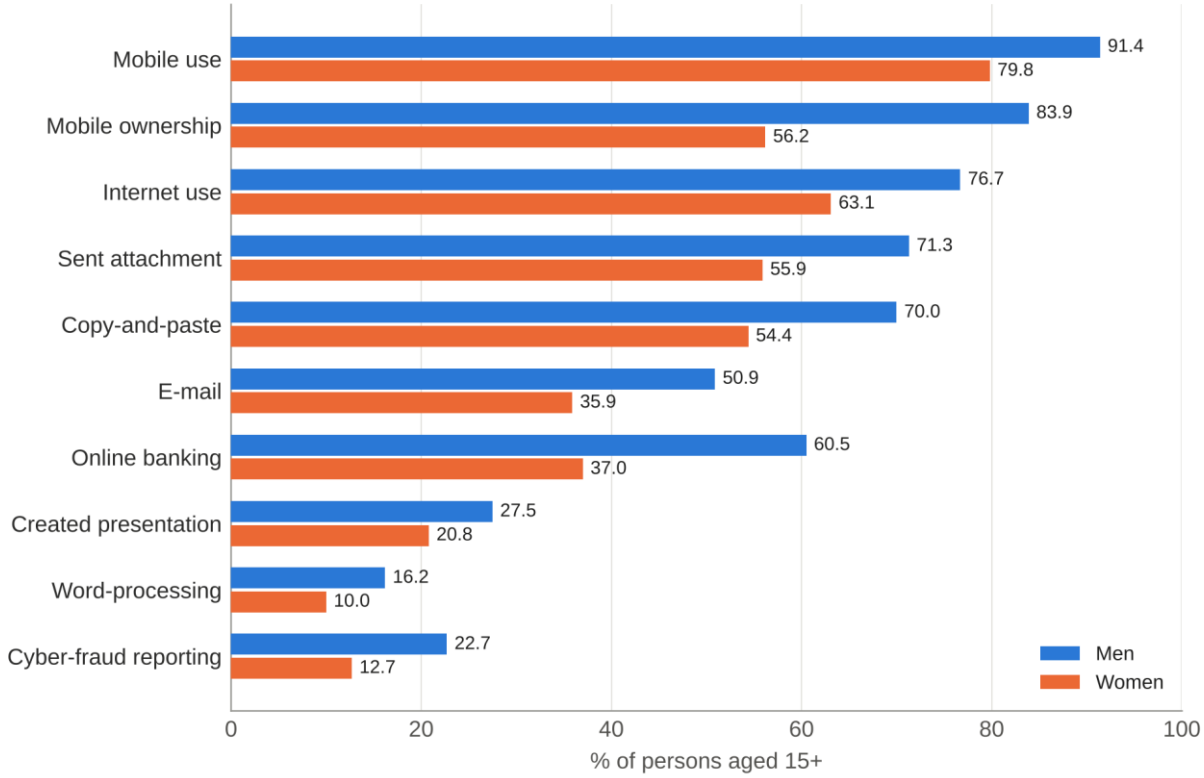
Gender gaps differ by indicator (Table 5, Figures 4–5). Among persons 15+, ownership shows the widest gap (27.7 pp; men 83.9%, women 56.2%), followed by online banking (23.5 pp). E-mail, copy-paste, attachments and internet use cluster at 13.6–15.6 pp, mobile use is 11.6 pp and the smallest gaps are for presentations (6.7) and word processing (6.2). In relative terms the ordering changes: the women/men ratio is lowest for cyber-fraud reporting (0.56), online banking (0.61) and word processing (0.62), so the small absolute gaps for advanced tasks reflect low prevalence rather than parity. Using the youth band, gaps in mobile use (2.0 pp) and internet use (4.6 pp) are small, but ownership (20.3 pp) and online banking (21.8 pp) persist: young women's internet use is 91.3% yet 57.5% report banking ability.

Table 5. Gender comparison, persons aged 15+, 2025

Indicator	Men (%)	Women (%)	Gap M–F (pp)	Women/men ratio	Odds ratio (W vs M)	Approx. 95% margin on gap (pp)
Mobile use (last 3 months)	91.4	79.8	11.6	0.87	0.37	±0.7
Mobile phone ownership	83.9	56.2	27.7	0.67	0.25	±1.4
Internet use (last 3 months)	76.7	63.1	13.6	0.82	0.52	±1.0
Sent message with attachment	71.3	55.9	15.4	0.78	0.51	n/a
Copy-and-paste	70.0	54.4	15.6	0.78	0.51	±1.4
E-mail ability	50.9	35.9	15.0	0.71	0.54	n/a
Online-banking ability	60.5	37.0	23.5	0.61	0.38	n/a
Created e-presentation	27.5	20.8	6.7	0.76	0.69	n/a
Word-processing ability	16.2	10.0	6.2	0.62	0.57	n/a
Cyber-fraud reporting ability	22.7	12.7	10.0	0.56	0.50	n/a

Source: NSO (2025), Tables 1–3, 6–12; RSE Tables 1–4; author's calculations. Odds ratio = [W/(100−W)] ÷ [M/(100−M)] from rounded percentages. Margins: 1.96 × SE of the gap, independence assumed (conservative); n/a = no RSE published.

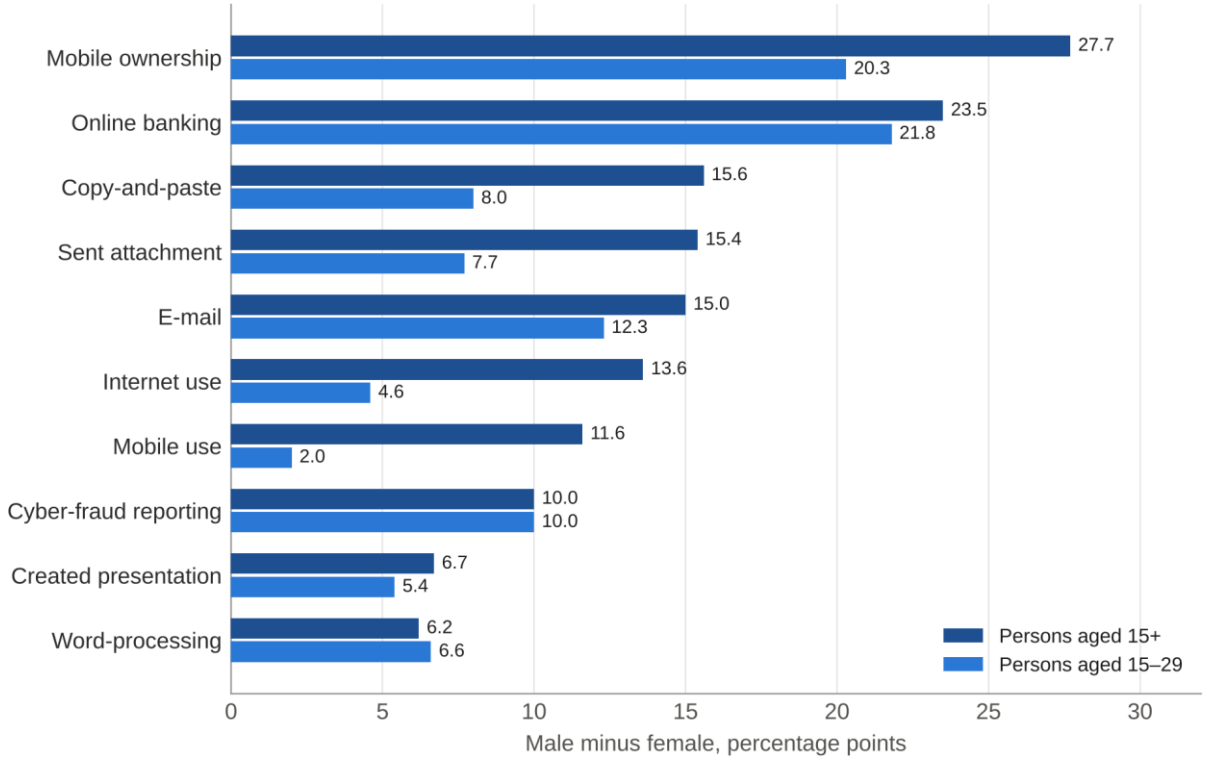
Figure 4. Digital adaptation of men and women aged 15+, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.

Key finding. Women trail men on every indicator, from 11.6 points in mobile use to 27.7 in ownership. **Implication.** The shortfall is smallest where a phone can be shared (use) and largest where personal control is required (ownership, banking), pointing to autonomy rather than mere availability.

Figure 5. Gender gap (male minus female) by indicator, persons aged 15+ and 15–29, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.
Note: gaps computed from published rounded percentages (±0.1 pp).

Key finding. Among 15–29-year-olds the gap shrinks sharply for use (2.0 pp, 4.6 pp) but only modestly for ownership (27.7 to 20.3) and banking (23.5 to 21.8); it is unchanged for cyber-fraud reporting (10.0). **Implication.** Generational turnover alone will not close ownership and capability gaps.

5.4 Rural–urban gaps

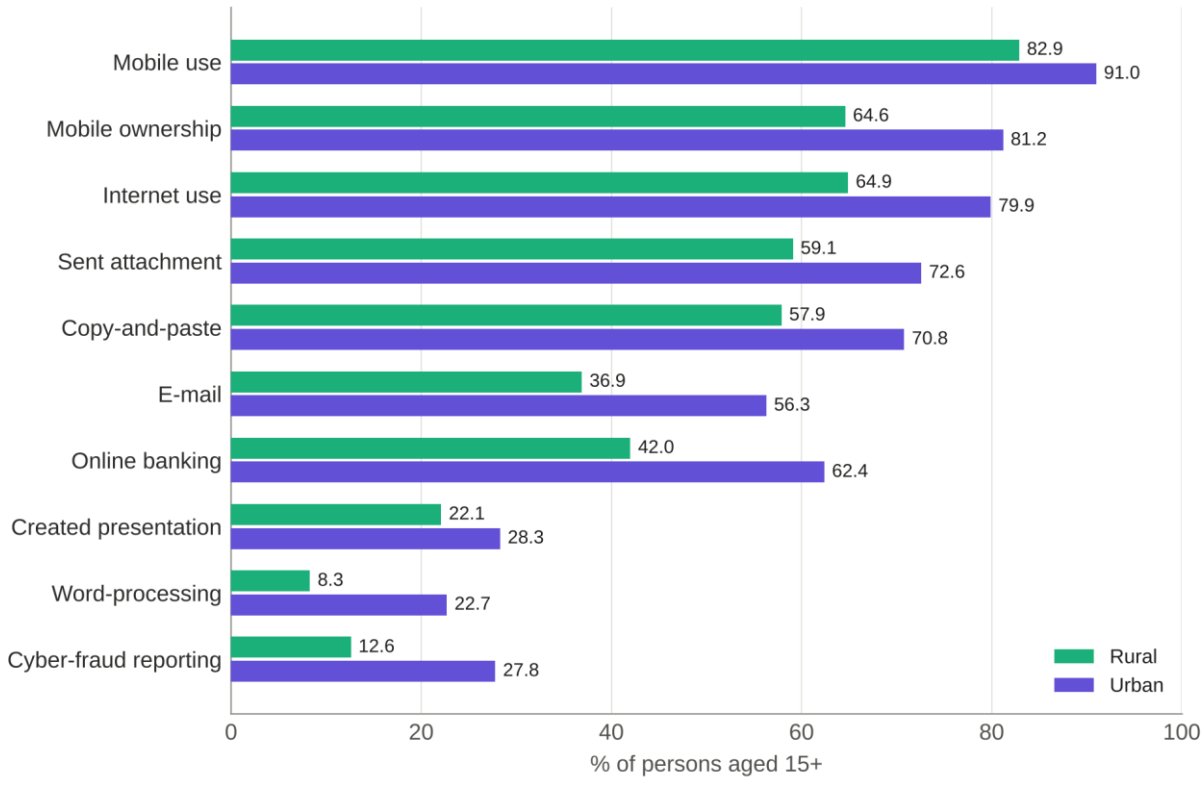
Among persons 15+, rural–urban gaps are largest for online banking (20.4 pp), e-mail (19.4), ownership (16.6), cyber-fraud reporting (15.2) and internet use (15.0), and smallest for presentations (6.2) (Table 6, Figures 6–7). The rural/urban ratio is lowest for word processing (0.37: 8.3% versus 22.7%) and cyber-fraud reporting (0.45). The youth band again separates use from capability: gaps are 0.8 pp for mobile use and 3.0 for internet use but 20.9 for word processing, 18.5 for cyber-fraud reporting and 16.6 for e-mail. For word processing and cyber-fraud reporting the youth gap exceeds the all-adult gap, so rural youth do not catch up on these tasks.

Table 6. Rural–urban comparison, persons aged 15+, 2025

Indicator	Rural (%)	Urban (%)	Gap U–R (pp)	Rural/urban ratio	Odds ratio (R vs U)	Approx. 95% margin on gap (pp)
Mobile use (last 3 months)	82.9	91.0	8.1	0.91	0.48	±0.8
Mobile phone ownership	64.6	81.2	16.6	0.80	0.42	±1.7
Internet use (last 3 months)	64.9	79.9	15.0	0.81	0.47	±1.2
Sent message with attachment	59.1	72.6	13.5	0.81	0.55	n/a
Copy-and-paste	57.9	70.8	12.9	0.82	0.57	±1.8
E-mail ability	36.9	56.3	19.4	0.66	0.45	n/a
Online-banking ability	42.0	62.4	20.4	0.67	0.44	n/a
Created e-presentation	22.1	28.3	6.2	0.78	0.72	n/a
Word-processing ability	8.3	22.7	14.4	0.37	0.31	n/a
Cyber-fraud reporting ability	12.6	27.8	15.2	0.45	0.37	n/a

Source: NSO (2025), Tables 1–3, 6–12; RSE Tables 1–4; author's calculations. Margins assume independent rural and urban strata; n/a = no RSE published.

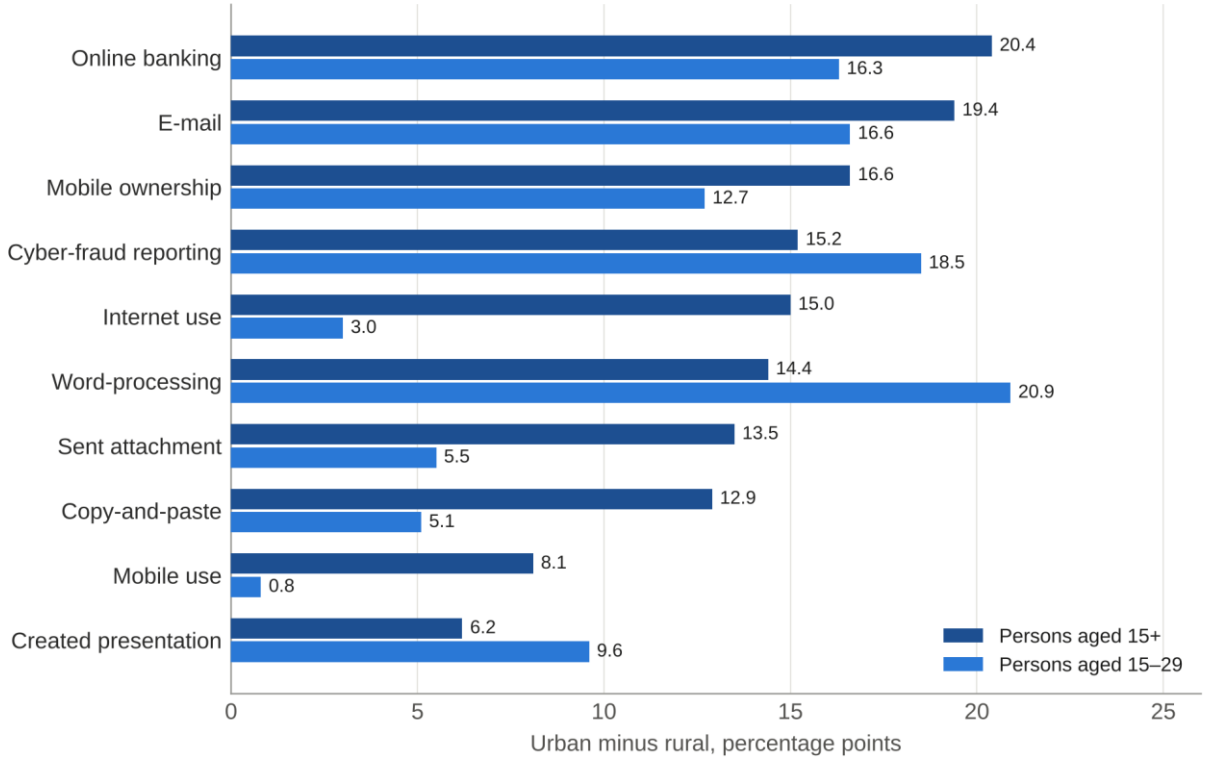
Figure 6. Digital adaptation of rural and urban persons aged 15+, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.

Key finding. Urban persons lead on all ten indicators; the contrast is largest in points for online banking and e-mail (36.9% versus 56.3%), and largest in proportion for word processing. **Implication.** Rural adults are connected but less equipped to use the connection for transactional and productive tasks.

Figure 7. Rural–urban gap (urban minus rural) by indicator, persons aged 15+ and 15–29, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.
Note: gaps computed from published rounded percentages (±0.1 pp).

Key finding. Rural–urban gaps for use shrink among youth, but gaps for word processing (20.9 pp), cyber-fraud reporting (18.5) and presentations rise. **Implication.** Sector disparity persists in higher-order skills even where use is saturated, which is where training and shared-computing access matter most.

5.5 Intersecting disadvantage

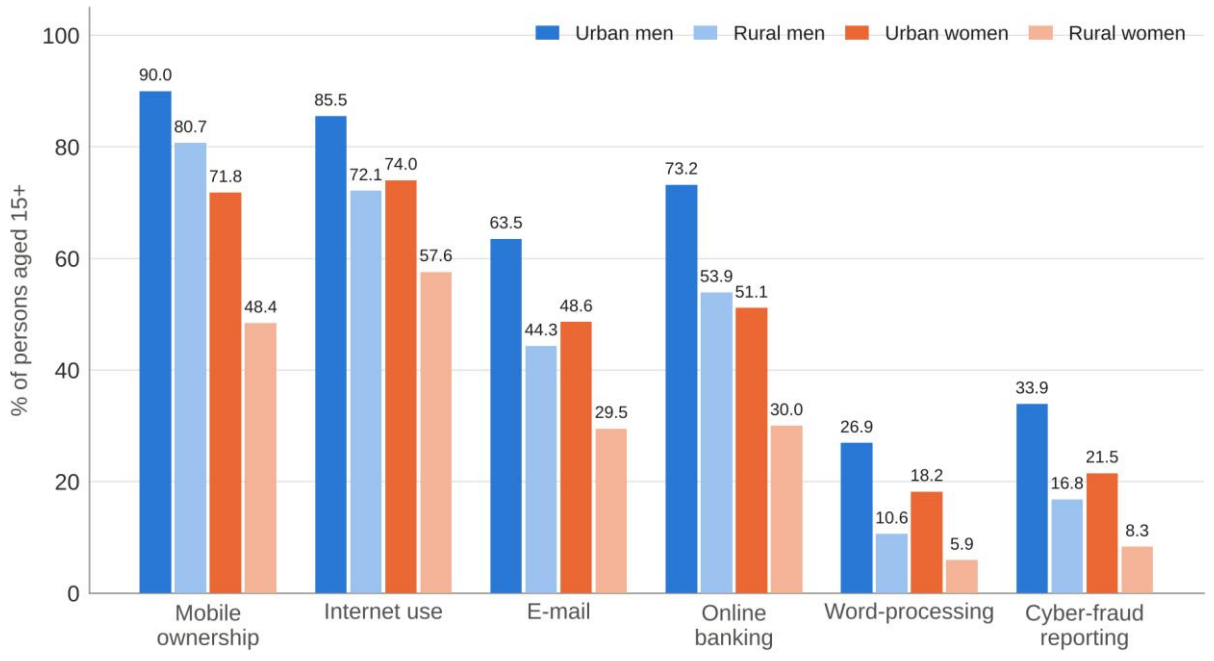
Combining gender and sector shows how disadvantages compound (Table 7, Figure 8). Rural women are furthest behind urban men on every indicator: 41.6 pp in ownership (48.4% versus 90.0%), 43.2 pp in online banking and 34.0 pp in e-mail. The gender gap in ownership is 32.3 pp in rural areas but 18.2 pp in urban areas, an interaction contrast of 14.1 pp. For word processing and cyber-fraud reporting the contrast is reversed (gender gaps are larger in urban areas), consistent with low rural prevalence among both sexes leaving little room for a gender difference. Urban women's internet use (74.0%) exceeds that of rural men (72.1%): for internet use, sector matters more than gender.

Table 7. Gender × sector, persons aged 15+, 2025 (per cent; gaps in percentage points)

Indicator	Urban men	Rural men	Urban women	Rural women	Urban men – rural women	Gender gap, rural	Gender gap, urban	Interaction (rural – urban)
Mobile use (last 3 months)	95.0	89.5	86.8	76.3	18.7	13.2	8.2	5.0
Mobile phone ownership	90.0	80.7	71.8	48.4	41.6	32.3	18.2	14.1
Internet use (last 3 months)	85.5	72.1	74.0	57.6	27.9	14.5	11.5	3.0
Sent message with attachment	79.1	67.2	65.8	50.9	28.2	16.3	13.3	3.0
Copy-and-paste	77.4	66.1	63.9	49.7	27.7	16.4	13.5	2.9
E-mail ability	63.5	44.3	48.6	29.5	34.0	14.8	14.9	-0.1
Online-banking ability	73.2	53.9	51.1	30.0	43.2	23.9	22.1	1.8
Created e-presentation	32.1	25.0	24.4	19.1	13.0	5.9	7.7	-1.8
Word-processing ability	26.9	10.6	18.2	5.9	21.0	4.7	8.7	-4.0
Cyber-fraud reporting ability	33.9	16.8	21.5	8.3	25.6	8.5	12.4	-3.9

Source: NSO (2025), Tables 1–3, 6–12; author's calculations from rounded published values. Interaction = rural gender gap minus urban gender gap; positive = gender gap larger in rural areas.

Figure 8. Digital adaptation by gender and sector, persons aged 15+, India, 2025



Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.

Key finding. Rural women are lowest on every indicator (48.4% own a phone; 5.9% report word-processing ability). Urban women exceed rural men on internet use, e-mail, word processing and cyber-fraud reporting, but not on ownership or banking. **Implication.** Sector shapes use and capability strongly, while gender shapes ownership; hypothesis H3 is supported for ownership only.

5.6 Age

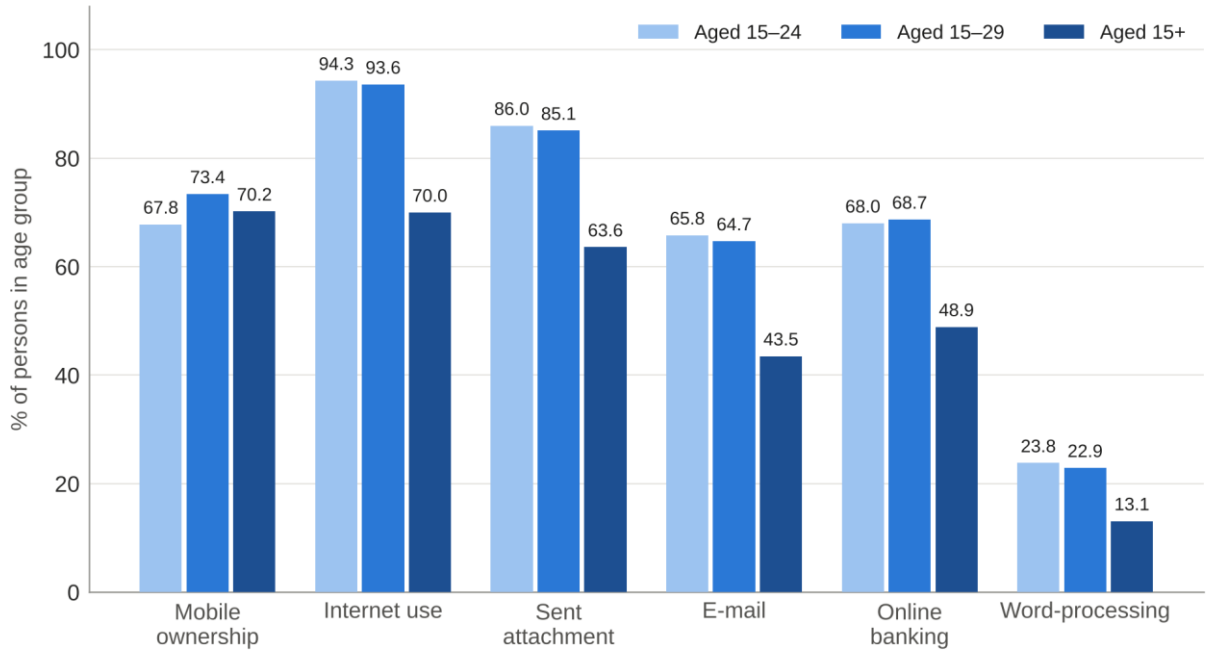
Table 8 and Figure 9 contrast youth with all adults. The 15–29 band reports far higher internet use (93.6% versus 70.0%), banking ability (68.7% versus 48.9%) and e-mail (64.7% versus 43.5%). Gender gaps fall with youth for most indicators but the reduction is uneven: the mobile-use gap falls from 11.6 to 2.0 pp, while the ownership gap falls only from 27.7 to 20.3 pp and the banking gap from 23.5 to 21.8 pp. Only age bands 15–24, 15–29 and 15+ are published; because 15+ includes 15–29, the contrast is indicative of an age gradient, not a clean comparison, and cannot show how capability declines beyond 30.

Table 8. Age gradient and gaps, all persons, 2025

Indicator	15–24 (%)	15–29 (%)	15+ (%)	Gender gap 15–29 (pp)	Gender gap 15+ (pp)	Rural–urban gap 15–29 (pp)	Rural–urban gap 15+ (pp)
Mobile use (last 3 months)	97.0	97.1	85.7	2.0	11.6	0.8	8.1
Mobile phone ownership	67.8	73.4	70.2	20.3	27.7	12.7	16.6
Internet use (last 3 months)	94.3	93.6	70.0	4.6	13.6	3.0	15.0
Sent message with attachment	86.0	85.1	63.6	7.7	15.4	5.5	13.5
Copy-and-paste	85.3	84.3	62.3	8.0	15.6	5.1	12.9
E-mail ability	65.8	64.7	43.5	12.3	15.0	16.6	19.4
Online-banking ability	68.0	68.7	48.9	21.8	23.5	16.3	20.4
Created e-presentation	32.6	32.2	24.2	5.4	6.7	9.6	6.2
Word-processing ability	23.8	22.9	13.1	6.6	6.2	20.9	14.4
Cyber-fraud reporting ability	26.3	26.9	17.7	10.0	10.0	18.5	15.2

Source: NSO (2025), Tables 1–3, 6–12; author's calculations. Age band 15+ includes 15–29.

Figure 9. Digital adaptation by age band, all persons, India, 2025



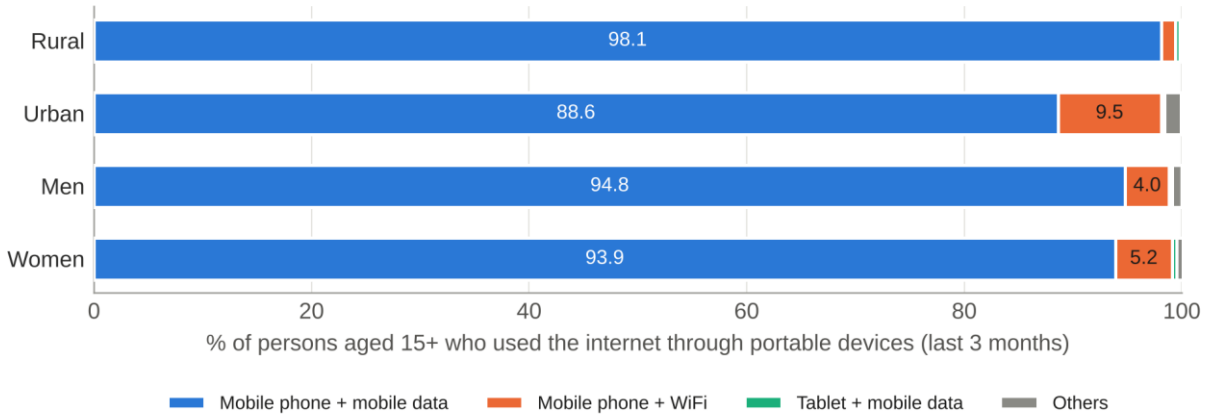
Source: NSO (2025), NSS Report No. 593, CMS: Telecom 2025 (NSS 80th Round, Jan–Mar 2025), Appendix A; author's calculations from the audited workbook.

Key finding. Internet use (94.3% at ages 15–24) and attachments (86.0%) are high among the young, but word processing is only 23.8% and ownership 67.8%. **Implication.** Even the youngest cohort is not uniformly capable; H2 is supported for use but only partly for ownership and advanced skills.

5.7 Devices and networks

Among persons 15+ who used the internet through portable devices, 94.4% mainly used a mobile phone on mobile data, 4.5% a phone on WiFi and under 1.5% a tablet or other device (Figure 10). Rural users rely almost exclusively on mobile data (98.1%); urban users use WiFi more (9.5% versus 1.3%). Men and women differ little (94.8% and 93.9%). Among phone owners aged 15+, 81.9% own a smartphone (smartphone-only plus both), 77.9% in rural and 88.0% in urban areas, so handset type is a smaller constraint than ownership itself. Computer use is not covered by the published device tables.

Figure 10. Main device and network used for internet access, persons aged 15+, India, 2025



Source: NSO (2025), Table 5 (Appendix A pp. A56–A63); author's calculations. Segments under 3% are unlabelled (rural WiFi 1.3%; tablet 0.3–0.4%; others 0.2–1.8%).

Key finding. Internet access is a mobile-data phenomenon in both sectors; urban WiFi is the main difference. **Implication.** The phone is the principal gateway, supporting the material-access argument that handset-only use may limit breadth of use, although the survey cannot test this.

5.8 Change since 2022–23

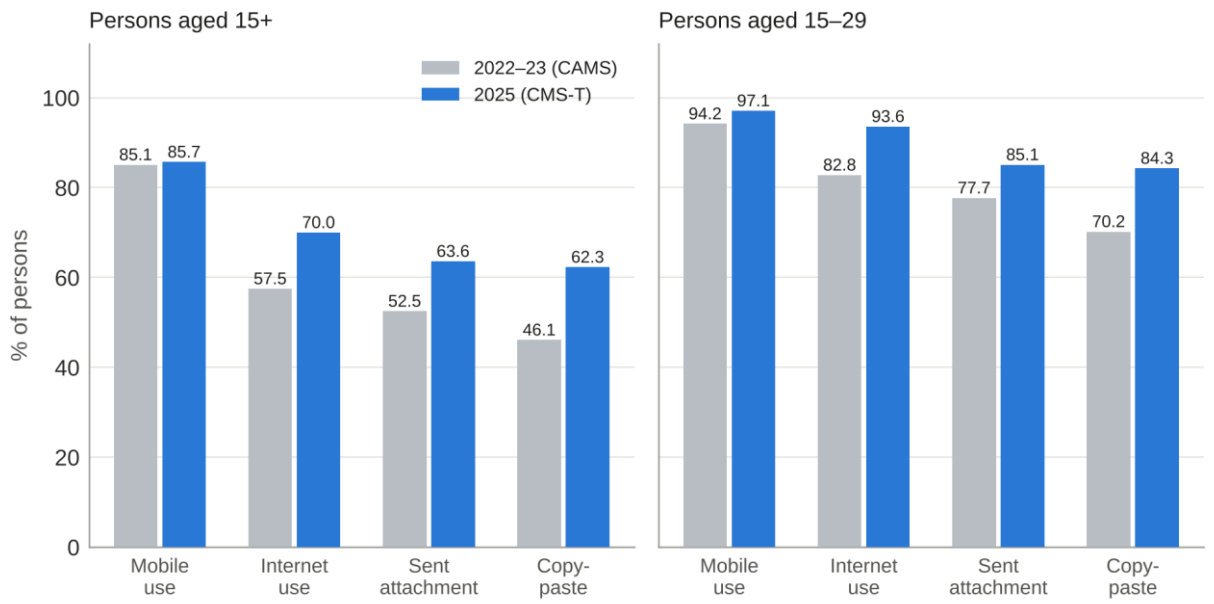
Four indicators have comparable wording and can be shown descriptively (Table 9, Figure 11). Mobile use among 15+ was almost unchanged (85.1% to 85.7%), consistent with saturation. Internet use rose from 57.5% to 70.0% (12.5 pp); women's use rose from 49.3% to 63.1% and men's from 65.5% to 76.7%. The gender gap in internet use among 15–29-year-olds narrowed from 11.0 to 4.6 pp, and the rural–urban gap among all adults from 20.8 to 15.0 pp. Rural women's internet use rose from 42.6% to 57.6%. E-mail is withheld because the 15+ and 15–29 changes are mutually inconsistent, and online banking because the question wording changed (NSO, 2025; workbook G21).

Table 9. Change in comparable indicators, 2022–23 to 2025 (all India)

Indicator, age	2022–23 (%)	2025 (%)	Change, all (pp)	Change, women (pp)	Change, men (pp)	Gender gap M–F (pp), 2022–23 → 2025
Mobile use, 15+	85.1	85.7	0.6	1.1	0.0	12.7 → 11.6
Mobile use, 15–29	94.2	97.1	2.9	4.2	1.6	4.6 → 2.0
Internet use, 15+	57.5	70.0	12.5	13.8	11.2	16.2 → 13.6
Internet use, 15–29	82.8	93.6	10.8	14.2	7.8	11.0 → 4.6
Sent attachment, 15+	52.5	63.6	11.1	12.1	10.3	17.2 → 15.4
Sent attachment, 15–29	77.7	85.1	7.4	10.0	5.0	12.7 → 7.7
Copy-and-paste, 15+	46.1	62.3	16.2	16.8	15.8	16.6 → 15.6
Copy-and-paste, 15–29	70.2	84.3	14.1	17.1	11.3	13.8 → 8.0

Source: NSO (2024), Tables 15, 17, 19, 20; NSO (2025), Tables 1, 3, 6, 7; audited workbook sheet D_Comparability. Descriptive differences between two surveys with different design, period and skill-question universe; no significance test is possible. Copy-and-paste is weaker (definition note absent in CAMS).

Figure 11. Comparable indicators in 2022–23 and 2025, persons aged 15+ and 15–29, India



Source: NSO (2024), Report 590, Tables 15, 17, 19, 20 (CAMS 2022–23); NSO (2025), Tables 1, 3, 6, 7; audited workbook sheet D_Comparability.
Note: descriptive comparison of two surveys with different design, period and universe filters; no significance test is possible.

Key finding. Use rose most where 2022–23 levels were lowest (internet 12.5 pp for 15+; 10.8 pp for 15–29), while mobile use barely moved. **Implication.** Growth is convergent but cannot be attributed to any programme; seasonal and design differences remain possible.

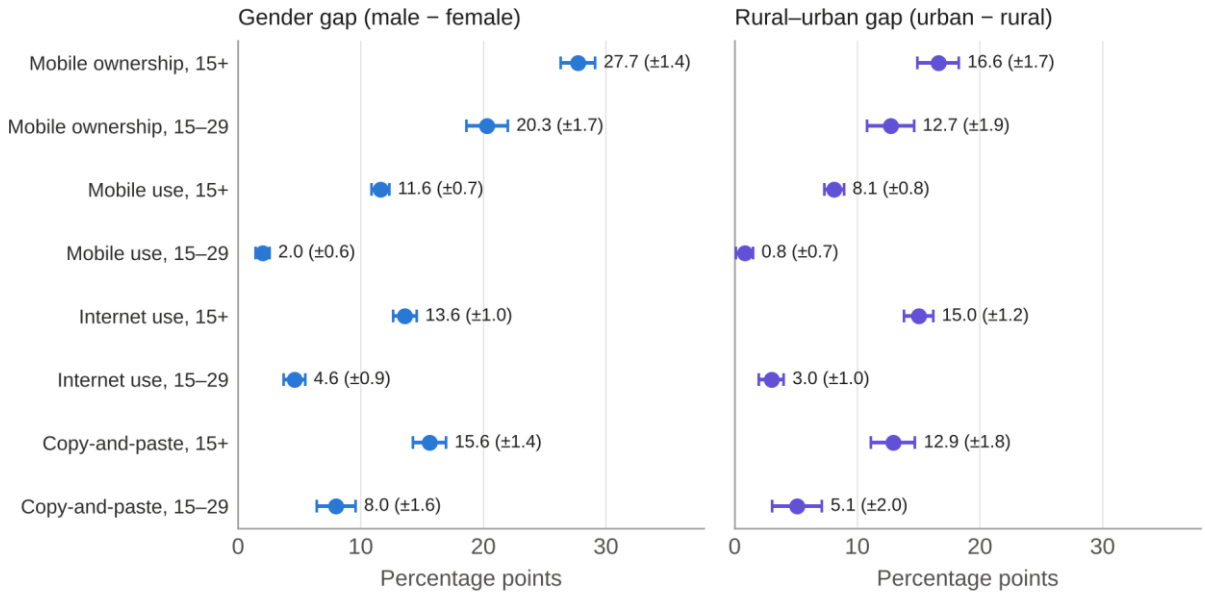
5.9 Statistical precision and inference

For the eight gender gaps and eight rural–urban gaps with published RSEs (four indicators, two age bands), none has an approximate 95% margin that includes zero (Tables 5–6, Figure 12). The smallest z-statistics are 6.9 for the gender gap and 2.3 for the rural–urban gap, both for mobile use among 15–29-year-olds (gaps of 2.0 and 0.8 pp); the latter is barely distinguishable from zero given rounding and unmodelled design effects. Margins on larger gaps are narrow (ownership gender gap 27.7 ± 1.4 pp), so the headline gaps are not sampling artefacts. For indicators without RSEs (attachments, e-mail, banking,

presentations, documents, cyber-fraud reporting) no margin can be given and those gaps are treated as descriptive.

On H4, across the ten indicators the gender gap is only weakly and non-significantly associated with the adoption level (Spearman $\rho = 0.56$, $p = 0.09$, $n = 10$), and the rural–urban gap is not ($\rho = -0.10$, $p = 0.78$). With ten non-independent indicators this cannot confirm or reject a diffusion curve; it shows only that gap size is not simply a function of prevalence.

Figure 12. Gender and rural–urban gaps with approximate 95% margins, four indicators with published RSEs



Source: gaps from NSO (2025) Tables 1–3, 7; RSEs from RSE Tables 1–4 (pp. A109–A120); author's calculations.
Whiskers: approximate 95% margin = 1.96 × SE of the gap; SE of gap = $\sqrt{(SE_1^2 + SE_2^2)}$, SE = RSE × estimate/100; independence assumed (conservative for gender gaps). Indicators without published RSEs are omitted.

Key finding. All sixteen gaps have margins that exclude zero, though the 15–29 rural–urban gap for mobile use (0.8 ± 0.7 pp) only just does. **Implication.** Gaps remain measurable even near saturation, so monitoring should continue to report them. Margins are conservative for gender gaps.

6. Discussion

The evidence answers the research questions in three parts. On RQ1, adaptation clearly extends beyond household connectivity to individual use, but only partly to ownership and functional capability. On RQ2 and RQ3, the size and location of gaps depend on the task: gender gaps are widest where personal control of a device or financial identity is required, rural–urban gaps are widest for transactional and productive tasks, and the two compound for rural women's ownership. On RQ4, internet use rose markedly, with narrowing gaps among youth, within the limits of comparability.

These patterns fit the literature. The dominance of mobile access (98.8% of internet households; 94.4% of portable-device users on phone data) supports van Deursen and van Dijk's (2019) claim that the first-level divide has shifted to material access, although CMS-T does not measure computer use or device quality. The gap between use and capability among youth illustrates DiMaggio and Hargittai's (2001) distinction between equipment and skill, and the 15.5-point use–ownership difference is a proxy for their autonomy dimension. Barboni et al. (2018) and GSMA (2025) also document large gender gaps in ownership and mobile internet use, and Roy et al. (2026) show that household wealth, household phone ownership and education account for most of the rural–urban internet gap among women, mechanisms

that CMS-T's published tables cannot test but that are consistent with the sector gradients in word processing and e-mail. In capability-approach terms (Sen, 1999; Kleine, 2013), a phone is a resource whose conversion into banking, communication and redress depends on skills, autonomy and context.

Absolute and relative gaps tell different stories. Gender gaps in points are smaller for advanced tasks than for ownership, yet women/men ratios are lowest for cyber-fraud reporting, online banking and word processing, so advanced capability is the most unequal in proportional terms. Table 10 synthesises the gaps. Two cautions apply. Self-reported capability is not tested proficiency, and associations in cross-sectional published aggregates must not be read causally. For example, the data cannot show that phone ownership causes higher banking ability or that rural location causes lower skills.

Table 10. Inclusion-gap summary, persons aged 15+ (ranked by urban men – rural women gap)

Indicator	Gender gap (pp)	Rural–urban gap (pp)	Urban men – rural women, 15+ (pp)	Urban men – rural women, 15–29 (pp)	Larger single dimension
Online-banking ability	23.5	20.4	43.2	35.6	Gender
Mobile phone ownership	27.7	16.6	41.6	30.5	Gender
E-mail ability	15.0	19.4	34.0	27.2	Sector
Sent message with attachment	15.4	13.5	28.2	12.4	Gender
Internet use (last 3 months)	13.6	15.0	27.9	7.1	Sector
Copy-and-paste	15.6	12.9	27.7	12.2	Gender
Cyber-fraud reporting ability	10.0	15.2	25.6	27.6	Sector
Word-processing ability	6.2	14.4	21.0	26.9	Sector
Mobile use (last 3 months)	11.6	8.1	18.7	2.5	Gender
Created e-presentation	6.7	6.2	13.0	14.8	Gender

Source: NSO (2025), Tables 1–3, 6–12; author's calculations from rounded published values (±0.1 pp).

7. Policy Recommendations

These recommendations follow from the interpretation above; they are not direct findings, and their effectiveness would need evaluation.

- **Prioritise personal ownership for women.** Ownership shows the widest gender gap (27.7 pp) while mobile use shows 11.6 pp, indicating widespread shared use. Programmes should address personal handsets and SIMs, affordability and household norms (Barboni et al., 2018), and monitor ownership by sex rather than household possession.
- **Move digital-literacy programmes from basic use to transactional and safety skills.** Use is near-universal among youth (93.6%), but banking (68.7%) and cyber-fraud reporting (26.9%) lag, with rural and female gaps of 18.5 and 10.0 pp in the latter. Curricula building on PMGDISHA (IANS, 2024) could add digital payments, grievance redress and document-based tasks.
- **Look beyond handset coverage in rural areas.** Rural households have 3.2% fibre and 9.1% of internet households have fixed or WiFi access, and rural word-processing ability is 8.3%. Shared community computing and assisted access could complement mobile coverage, subject to evaluation.

- **Address non-use at household level.** 39.2% of households without internet cite not knowing how to use it and 21.3% cite lack of need; assisted demonstration and locally relevant services may matter as much as network expansion.
- **Strengthen the evidence base.** NSO should publish RSEs for all indicators, state the denominators of skill indicators, cross-tabulate by education, consumption quintile and social group, and release unit-level data so that drivers can be modelled and State-level precision assessed. Affordability and quality of connectivity should be added to the schedule.

8. Limitations

CMS-T is cross-sectional and self-reported, so reported capability is not demonstrated proficiency and no causal claim follows. The analysis uses published, rounded, design-weighted aggregates rather than unit-level data, so no multivariable adjustment, weight re-estimation or exact variance estimation was possible. The base of the skill percentages is not stated, execution and ability indicators differ in concept, and RSEs exist for five indicators only. Education, income, social group, disability and State/UT analysis are unavailable. Household and individual indicators are not linked. The 2022–23 comparison involves different designs, periods and universes and excludes e-mail and online banking. The ITU target values come from a mirrored, undated copy and should be confirmed on itu.int. Scheerder et al. (2017) was read at abstract level only. One audit check (household network coverage) did not close, and 'person' includes transgender persons without a separate estimate.

9. Conclusion

Digital adaptation in India has moved well beyond household connectivity to individual use, almost completely among the young, but functional capability and personal ownership lag, especially for women and rural residents. Gender gaps are widest in ownership (27.7 pp) and online banking (23.5 pp); rural–urban gaps are widest in e-mail, banking and word processing; and rural women are furthest behind on every indicator. Use has risen quickly since 2022–23, but use is no longer the binding constraint. The agenda is therefore ownership, skills for transactions and safety, and better data on who is left behind and why.

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Appendix A. Data audit checks

Table 11 lists the thirteen validation checks run on the audited workbook before any statistic was reported. Each check was computed independently in code from the workbook cells; "n" is the number of cells or rows tested.

Table 11. Data audit checks and results

#	Check	Passed / tested	Result	Detail
1	Completeness of 270-cell estimate grid	270/270	Pass	No missing cells
2	All percentages within 0–100	270/270	Pass	Range 5.9–98.0
3	Person estimate between male and female	90/90	Pass	No violations (±0.05)
4	All-India estimate between rural and urban	90/90	Pass	No violations (±0.05)
5	Gaps recomputed vs workbook formulas	180/180	Pass	Max difference <1e-13
6	Evidence matrix vs source sheet (C_Data)	100/100	Pass	15+ and 15–29, 10 indicators
7	Distribution rows (Tables 1.1, 2.1, 4, 12.1) sum to 100	108/108	Pass	Max deviation 0.10 pp
8	Device/network distribution (Table 5) sums to 100	27/27	Pass	Max deviation 0.20 pp
9	Household mobile = smartphone-only + other-only + both	3/3	Pass	Reproduces Highlights 94.4/96.6
10	Households with + without internet = 100	3/3	Pass	Sums exactly 100.0
11	Reasons for no household internet sum to 100	3/3	Pass	Sums 100.0–100.1 (rounding)
12	Network types: mobile + fixed – both = 100	0/3	FAIL	Coverage 99.1–99.4%; residual 0.6–0.9 pp unexplained
13	Sample totals: FSUs and persons add to published totals	2/2	Pass	2,395 + 1,987 = 4,382; 82,573 + 59,492 = 142,065

Source: audited workbook (sheets B_Evidence, C_Data, C2_Household, C3_Distributions, E_Reliability); author's checks. Pseudo-weight check: implied male share of 15+ = 0.500–0.509 and implied rural share = 0.654–0.667 across ten indicators (see Section 4.3).

Appendix B. Reliability and sample precision

Table 12 shows how little precision is available below all-India level. Relative standard errors exist only for mobile use, ownership, internet use, copy-paste and household internet, and several States/UTs have very few sampled units. State/UT rankings are therefore not supportable for any indicator.

Table 12. Precision of published RSEs and smallest State/UT samples

Item	Age / FSUs	Cells / FSUs	Cells RSE > 25% / households	Cells RSE > 50% / households	States/UTs with RSE > 25% / persons
RSE_Table 1 (mobile use)	15-24	330	2	0	none
RSE_Table 1 (mobile use)	15-29	330	1	0	none
RSE_Table 1 (mobile use)	15 and above	330	1	0	none
RSE_Table 2 (ownership; titled smartphone)	15-24	330	10	1	none
RSE_Table 2 (ownership; titled smartphone)	15-29	330	3	0	none
RSE_Table 2 (ownership; titled smartphone)	15 and above	330	1	0	none
RSE_Table 3 (internet use)	15-24	330	2	1	none
RSE_Table 3 (internet use)	15-29	330	2	0	none
RSE_Table 3 (internet use)	15 and above	330	2	0	none
RSE_Table 4 (copy-paste)	15-24	330	12	2	Goa, Ladakh
RSE_Table 4 (copy-paste)	15-29	330	16	6	Ladakh
RSE_Table 4 (copy-paste)	15 and above	330	19	7	Goa, Ladakh
RSE_Table 5 (household internet facility)	households	108	0	0	none
Panel B: smallest State/UT samples (Table 16)	**Rural FSUs**	**Urban FSUs**	**Rural households**	**Urban households**	**Persons, rural / urban (all ages)**
Ladakh	2	4	16	32	86 / 147
Lakshadweep	2	4	16	32	68 / 106
Goa	4	4	32	32	122 / 94
Puducherry	4	8	32	64	112 / 209
A & N Islands	6	5	48	40	160 / 107
Chandigarh	no sample	6	no sample	48	no sample / 122
Dadra & Nagar Haveli & Daman & Diu	6	6	48	48	180 / 200

Source: NSO (2025), RSE Tables 1–5 (pp. A109–A121) and Table 16 (p. A107); audited workbook sheet E_Reliability. Degenerate RSEs of 0.00 occur where all sampled persons answered alike (e.g., Lakshadweep) and do not indicate high precision. "No sample" = no rural sample in Chandigarh.

Appendix C. Data availability against the brief

- **Access, device ownership, internet usage:** available at all-India level by sex, sector and three age bands (15–24, 15–29, 15+).
- **Gender and rural–urban comparisons:** available for all ten individual indicators; household indicators have no gender cut.
- **Education:** collected in the schedule but not tabulated in the published telecom tables; not analysed.

- **Socio-economic groups (consumption, caste/religion) and disability:** not tabulated; not analysed. External evidence (Roy et al., 2026) is used as context only.
- **Regions:** State/UT tables exist but the survey was not designed for State estimates; RSEs exist for five indicators only and some States have two to six sampled rural FSUs. Region is therefore the rural–urban sector.
- **Survey weights:** published estimates are already design-weighted; unit-level data were not available, so weights were audited by implied pseudo-weights rather than re-applied.
- **Sample sizes and missing values:** FSU, household and person counts exist for all-India and State/UT (all ages) only. The 270-cell indicator grid has no missing cells; blank cells occur only where Table 3.1 has no sample observations.

Appendix D. Reproducibility notes

Gaps are computed as male minus female and urban minus rural from the rounded published percentages (each accurate to ±0.1 pp). Parity ratio = female/male or rural/urban; odds ratio = $[p_1/(100-p_1)] \div [p_2/(100-p_2)]$. Approximate standard error of an estimate = RSE × estimate/100; SE of a gap = $\sqrt{(SE_1^2 + SE_2^2)}$; 95% margin = 1.96 × SE. Interaction contrast = rural gender gap minus urban gender gap. Spearman correlations use the ten individual indicators for persons 15+. The companion workbook (digital_adaptation_calculations.xlsx) contains every computed statistic reported here, and the figures were drawn from the same values.